



ANALYTICAL SOLUTIONS TO PREVENT APPARENT SNAP-BACK IN THE MODELING OF ELASTIC-DAMAGING PROCESSES

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Introduction

The presence of microdefects within the structure of a brittle solid plays a dominant role in the manner in which the material responds to externally applied stresses. Equally important are the type, size, distribution, evolution, and interaction of microdefects as documented by electron microscopy, acoustic emission techniques, laser interferometry, and other nondestructive tests. When a loading state involves a confining pressure of a sufficient magnitude, such that nucleation of new microdefects and the propagation of existing ones are inhibited, brittle solids display a ductile response. When the same materials are stressed in a low pressure regime, responses become very brittle with little or no ductility.

In problems where the nonlinear response of a solid arises as a consequence of the gradual evolution of crack-like defects, micromechanically-based phenomenological models have become viable predictive tools ([1],[2],[3],[4],[5],[6]). The success of the approach lies in its ability to relate directly to the physics of the problem in the same manner that plasticity connects with slip-like deformations in metals. At the same time, the approach strikes a balance between theoretical rigor and computational feasibility.

The theory is cast within the general setting of irreversible thermodynamics with internal variables under the assumption of the existence of a constrained equilibrium state. This assumption enables one to define the state of a solid based on the current values of the state variables. Kinetic relations for fluxes are subsequently postulated, guided by the defect evolution processes at the mesoscale. In this way, an approach is formulated, bypassing the extensive data bank requirements of the micromechanical damage theories ([7],[8],[9]). In the latter theories, damage modeled at the mesoscale is mapped on the macroscale by proper homogenization techniques. The computational efforts to handle micromechanical models have become extensive beyond our current capabilities in engineering analysis. This paper takes the micromechanically-based macromechanical theory approach.

During the process of damage, material bonds are severed, leading to the alteration of elastic properties. It is, therefore, plausible to employ the changes in compliance or stiffness as measures of damage as they relate directly to the physics of the problem. A three-dimensional extension of this concept was the novel approach of Ortiz [3], in which the fourth order compliance tensor was used as an internal variable. In this way, the directionality of damage was accounted for, enabling one to address load-induced anisotropy in brittle solids. Ortiz's model was further improved by Yazdani and Schreyer [10] and later by Yazdani [11].

Recently, Rasset et al. [12] reported that a problem arose when a bilinear damage function was used instead of the logarithmic one that was utilized in the earlier models. Their work indicated that an apparent snap-back in the solution, for the uniaxial tensile path, was observed in the strain softening regime. However, no strategy for dealing with that problem was outlined in their paper.

Bilinear functions are of considerable use in nonlinear constitutive modeling, particularly by practitioners who wish to calibrate their computational codes and have trial runs. It is, therefore, the intent of this paper to explore the issue further and propose a strategy by which the problem of the apparent snap-back in strain softening, for the same loading path, is removed. It should be emphasized that some physical damage processes do lead to localized deformations and possible snap-back ([13]). In this paper, the term *apparent* snap-back refers to the development of an internal contradiction in the damage model where snap-back in the deformation is predicted even though the theory was not structured to capture this aspect of material response. The paper will start with a review of the fundamental concepts based on the laws of thermodynamics, will consider anisotropic damage formulation, assuming elastic damage processes, and will propose a simple perturbation to the solution of a constrained ordinary differential inequality (ODI) by which the internal contradiction of the solution is avoided. Finally, the paper will illustrate the model for the uniaxial tensile stress path.

General formulation

Assume that a nonpolar continuum approach can effectively be taken for the rate-independent modeling of quasi-brittle solids in which only mechanical deformations are considered. Adopting a stress space formulation within the context of the internal variable theory of thermodynamics ([14],[15],[16],[17]), the Clausius-Duhem inequality yields:

$$\dot{G} - \dot{\sigma} : \epsilon \geq 0, \quad (1)$$

in which $G(\sigma, k)$ represents the Gibbs free energy (GFE) as a state function. The state variables are given by the Cauchy stress tensor, σ , and the cumulative damage parameter, k . The inequality, given by Equation (1), must be satisfied for all admissible processes in which the strain tensor is denoted by ϵ . The symbol ":" indicates the tensor contraction operation, and superdots are time rates. For small deformations caused by elastic-damaging processes, the total strain can be given by the following constitutive relation as:

$$\epsilon(\sigma, k) = C(k) : \sigma, \quad (2)$$

in which $\mathbf{C}$ represents the fourth order compliance tensor for the material. The dependence of $\mathbf{C}$ on k reflects the idea that damage alters elastic properties ([18],[19]) and allows for the description of load-induced anisotropy. The use of a fourth order tensor to record damage and its directionality has gained a wide acceptance among researchers in the field (see the review article by Krajcinovic [1]). The structure of Equation (1) establishes two fundamental results. The first one is the dissipation inequality, given as:

$$d_s = \frac{\partial G}{\partial k} \dot{k} \geq 0, \quad (3)$$

and the second one is the statement that GFE is a potential for the strain tensor as:

$$\boldsymbol{\varepsilon}(\boldsymbol{\sigma}, k) = \frac{\partial G(\boldsymbol{\sigma}, k)}{\partial \boldsymbol{\sigma}} = \mathbf{C}(k) : \boldsymbol{\sigma}. \quad (4)$$

Equation (4) when integrated with respect to $\boldsymbol{\sigma}$ yields the general form of the GFE as:

$$G(\boldsymbol{\sigma}, k) = \frac{1}{2} \boldsymbol{\sigma} : \mathbf{C}(k) : \boldsymbol{\sigma} + g(k), \quad (5)$$

in which $g(k)$ is a non-negative scalar function associated with the surface energy of defect generation and growth. Denoting $\mathbf{C}^0$ as the compliance tensor for an undamaged state and letting $\mathbf{C}^c$ be the added flexibility tensor due to damage, the total compliance tensor is written as $\mathbf{C}(k) = \mathbf{C}^0 + \mathbf{C}^c(k)$. The substitution of Equation (5), together with the expression for $\mathbf{C}(k)$, into Equation (3) results in the following alternate form of the dissipation inequality in terms of state fluxes, $\dot{\mathbf{C}}^c$ and $\dot{g}$, as:

$$d_s = \frac{1}{2} \boldsymbol{\sigma} : \dot{\mathbf{C}}^c : \boldsymbol{\sigma} + \dot{g} \geq 0. \quad (6)$$

Suppose that fluxes are expressed in a linear form in k as:

$$\dot{\mathbf{C}}^c(k) = \dot{k} \mathbf{R}, \quad \dot{g}(k) = \dot{k} f, \quad (7)$$

in which $\mathbf{R}$ is the damage response tensor that determines the direction of elastic damage processes and where f is a scalar function. A specific form of the response tensor will be given in the next section. It will suffice here to state that if $\mathbf{R}$ is proportional to the fourth order identity tensor, the formulation will correspond to isotropic damage. If the directionality of damage is accounted for in the choice of $\mathbf{R}$, the model would allow for the description of anisotropic processes. The ease and the generality of such a formulation are quite appealing and will be explored further in the present paper for the tensile stress states.

To progress further, a damage criterion is needed. This is accomplished by defining a damage potential, $\Psi(\boldsymbol{\sigma}, k)$, such that the condition $\Psi=0$ defines a damage surface encompassing an elastic

domain wherein $\Psi < 0$. For rate independent processes, the condition $\Psi > 0$ is excluded and not allowed. The loading-unloading statement is made using the standard Kuhn-Tucker form as:

$$\Psi \leq 0, \quad \dot{k} \geq 0, \quad \dot{k}\Psi = 0, \quad (8)$$

in which Equation (8) represents a set of equations that must be satisfied simultaneously for all admissible processes. The general form of the damage surface is obtained by substituting Equation (7) into Equation (6) and introducing a scalar function, $\tilde{g}^2(\sigma, k)$, as:

$$\Psi(\sigma, k) = \frac{1}{2} \sigma : \mathbf{R} : \sigma + f - \tilde{g}^2(\sigma, k) = 0. \quad (9)$$

Note that the physical identification of individual functions f or $\tilde{g}$ is not necessary as long as the sum of the two functions, f and $\tilde{g}^2$, called the damage function, $t(\sigma, k)$, and defined by $t^2 = 2(\tilde{g}^2 - f)$, can be experimentally identified. The determination of t is described below.

Anisotropic Damage in Tension

To facilitate the discussions below, let the cross composition operation of tensors be denoted by the symbol " $\hat{\otimes}$ ", such that $\mathbf{U} = \mathbf{V} \hat{\otimes} \mathbf{W} = V_{ik} W_{jl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l$. The symbol " $\otimes$ " designates the tensor product operation, and $\hat{\mathbf{e}}_i$ are the orthonormal base vectors. The stress tensor may be defined by eigenvectors, $\mathbf{q}_a$, associated with eigen stresses, σ_a , as $\sigma = \sum_{a=1}^3 \sigma_a \mathbf{q}_{(a)} \otimes \mathbf{q}_{(a)}$. Employing the Heaviside function, $H(\cdot)$, let the positive spectral operator for the stress tensor be denoted by $\mathbf{Q}^+$ and defined by:

$$\mathbf{Q}^+ = \sum_{a=1}^3 H(\sigma_a) \mathbf{q}_{(a)} \otimes \mathbf{q}_{(a)}. \quad (10)$$

The cross composition of the tensor $\mathbf{Q}^+$ by itself yields a fourth order tensor operator, $\mathbf{P}^+ = \mathbf{Q}^+ \hat{\otimes} \mathbf{Q}^+$, such that when $\mathbf{P}^+$ is applied to the stress tensor, σ , in the principal basis, the stress tensor is mapped into its positive cone. The positive cone is defined as a space wherein all negative eigenvalues of the stress tensor are removed. This operation is given symbolically as $\sigma^+ = \mathbf{P}^+ : \sigma$. Alternately, the same procedure can be shown using the general transformation tensor, $\mathbf{Q}$, operating between base vectors, $\mathbf{q}_a$ and $\hat{\mathbf{e}}_i$, such that $\mathbf{Q} = \mathbf{q} \otimes \hat{\mathbf{e}}$. The positive cone of the stress tensor is then obtained as $\sigma^+ = [\mathbf{Q}^+ \cdot \mathbf{Q}] \cdot \sigma \cdot [\mathbf{Q}^+ \cdot \mathbf{Q}]^T$, where the superscript "T" designates the transpose operation. See also [2], [3], and [5] for similar operational procedures. With these preliminaries and assuming that penny-shaped cracks are the dominant microdefects present in brittle solids, the necessary conditions for crack opening in *mode I* damage (cleavage cracking) can formally be stated as follows.

Let a planar crack be defined by its radius, r , and its normal vector to the plane of the crack as $\mathbf{n}$. The crack is said to be open (not necessarily active) if i) $\sigma^+ \neq \mathbf{0}$, and ii) $\mathbf{n} \cdot \sigma^+ \cdot \mathbf{n} \neq 0$. The

conditions i and ii above constitute the necessary conditions for crack opening but are not sufficient to characterize cracks as being active or passive. A crack is said to be active or operational if $\dot{t} \neq 0$. For this latter case, the necessary conditions are the same as statements i and ii above, and the sufficiency conditions come from the loading statement of the damage criterion given by Equation (8).

This last statement should be viewed with caution. Within a continuum formulation where smearing formalism is employed, there may well be individual microcracks that remain inoperational or passive while the necessary and sufficient conditions for active microcracks are met. This scenario may arise due to microcracks' interaction, whereby some cracks are shielded from the effects of far field stresses by some other active crack arrays. Therefore, the stated conditions on individual crack opening and propagation should not be interpreted as the general one-to-one relation between meso and macro mechanics of crack operation. Rather, statements should be regarded as an indication that similarity of approaches exists between mesomechanical modeling and micromechanically-based phenomenological theories when energy dissipation, due to active microcracks, is considered. In situations where microcrack interaction is not pronounced (distributed, uniform, and dilute density of cracks within a representative volume element), the necessary and sufficient conditions as stated above would hold true for individual cracks as well.

To bring these statements into the general formulation, consider the damage response tensors, $\mathbf{R}$ and $\mathbf{M}$, as (see [11]):

$$\mathbf{R} = \frac{\boldsymbol{\sigma}^+ \otimes \boldsymbol{\sigma}^+}{\boldsymbol{\sigma}^+ : \boldsymbol{\sigma}^+} + \gamma \Phi(\mathbf{I} - \mathbf{i} \otimes \mathbf{i}), \quad (11)$$

in which γ is a material parameter associated with the strength characteristics of the material and where $\mathbf{I}$ and $\mathbf{i}$ represent fourth and second order identity tensors, respectively. The scalar function, Φ , is used for convenience and is defined as $\Phi = H(\lambda)$, where λ is the maximum eigenvalue of $\boldsymbol{\sigma}^+$. Note also that Equation (11) is structured in such a way that under purely compressive stresses, the response tensor $\mathbf{R}$ becomes a null tensor and does not contribute to the deformation processes.

Uniaxial tension

The substitution of Equation (11) into Equation (9) establishes the following relation between the uniaxial stress, σ_1 , and the damage function as:

$$\sigma_1^2 = t^2(\sigma, k). \quad (12)$$

Suppose that the damage function t is given as $t(\sigma, k) = A(\sigma)\Gamma(k)$, where $A(\sigma)$, known as the damage strength function as given in [11], reduces to $A(\sigma) = f_t$ for the uniaxial tensile path, with f_t being the uniaxial tensile strength of the material. For a bilinear damage function, $\Gamma(k)$ is expressed as:

$$\Gamma(k) = \begin{cases} (1/k^*) k & \text{for } k \leq k^* \\ 1 - \eta (k - k^*) & \text{for } k \geq k^*, \end{cases} \quad (13)$$

where k^* reflects the damage corresponding to the maximum value of $\Gamma(k)$ and where η , a positive scalar, defines the slope of the damage function in the strain softening region. Using the rate form of Equation (2) together with Equations (7), (9), and (11), and integrating over the path, the closed form of the stress-strain relation for the uniaxial tensile stress path can be shown to be:

$$\varepsilon_1 = \left\{ \frac{1}{E_0} + k \right\} \sigma_1, \quad (14)$$

in which E_0 represents the initial Young's modulus for the material. This is the same result reported in [12]. A further insight into the structure of Equation (14) reveals that the strain, ε_1 , is not a monotonically increasing function of the damage parameter, k . This feature is illustrated in Figure 1 for $k^* = (e-1)/E_0$ in which e is the natural number. For convenience and for illustration purposes E_0 and f_t are both taken as one, and ε_u , which is the strain corresponding to the maximum stress, is computed from Equation 14. It is shown that no matter how steep or mild the slope of the bilinear damage function in the post peak regime is, the apparent snap-back is inevitable. With a larger η (steeper descending curve), the snap-back takes place earlier than when η is small (milder). The following solution strategy pertains to the strain softening regime strictly. No snap-back problem exists in the strain hardening region.

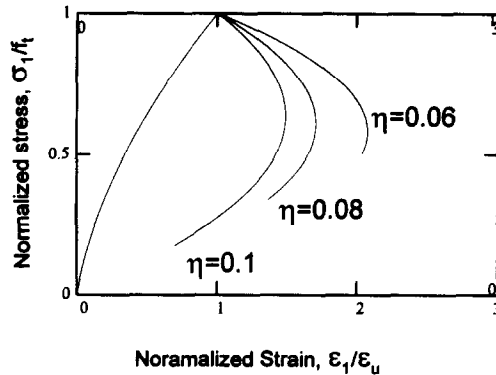


Figure 1. Theoretical stress-strain curve in uniaxial tension using Equation (13).

It is clear that for the strain function to monotonically increase with increase in damage, the slope, η , must change. Indeed, one can attempt to develop an evolutionary expression for η based on constraints that $\varepsilon_{1,k} > 0$ and $\Gamma_{,k} < 0$, where subscripts $_{,}$ signify differentiation with respect to the parameters that follow. In this paper, an expression for $\Gamma(k)$ based on the solution to a constrained ODI is derived. In this way, we bypass potential problems associated with postulating evolutionary expressions. Let the damage function be given by:

$$\Gamma(k) = \begin{cases} k / k^* & \text{for } k \leq k^* \\ \bar{\Gamma}(k) & \text{for } k \geq k^*, \end{cases} \quad (15)$$

where we are looking to find a particular form of $\bar{\Gamma}(k)$ in which the inequalities $\epsilon_{i,k} > 0$ and $\bar{\Gamma}_{,k} < 0$ are satisfied. Differentiating Equation (14) with respect to the damage parameter and using Equations (12) and (15) yield the following constrained ODI that must be satisfied:

$$\bar{\Gamma}(k) + \left(\frac{1}{E_0} + k\right) \bar{\Gamma}_{,k}(k) > 0, \quad (16)$$

subject to $\bar{\Gamma}_{,k} < 0$. To solve the inequality relation of Equation (16), consider a complimentary ordinary differential equation (ODE) given as $\bar{\Gamma}^c(k) + \left(\frac{1}{E_0} + k\right) \bar{\Gamma}^c_{,k}(k) = 0$. The solution to this first order and linear ODE is given as:

$$\bar{\Gamma}^c = \frac{c_1}{\frac{1}{E_0} + k}, \quad (17)$$

where the constant of integration, c_1 , is determined from the initial condition. Now consider the solution to the inequality (16) in the form of $\bar{\Gamma}(k) = \bar{\Gamma}^c(k) + \Theta(k)$, where $\Theta(k)$ is regarded as the perturbation to the complimentary solution. The judicious choice of $\Theta(k)$ is made by the analysts. Here, we propose $\Theta(k)$ to be a scalar, c_2 , for convenience. To evaluate the constant c_2 , let $\lim(\Gamma(k))$ be Γ^* as $k \rightarrow \infty$. This yields that c_2 is equal to the limit value Γ^* , where Γ^* can be obtained from experimental observation on the uniaxial stress-strain curves. See [20], for example, for typical stress strain curves in uniaxial tension for mortar and concrete specimens. To evaluate c_1 , the initial condition for $\Gamma(k)$ is used, yielding $c_1 = (1 - \Gamma^*) \left(\frac{1}{E_0} + k^*\right)$. The final form of $\bar{\Gamma}$ is given as:

$$\bar{\Gamma} = \frac{\left(\frac{1}{E_0} + k^*\right)(1 - \Gamma^*)}{\frac{1}{E_0} + k} + \Gamma^* \quad (18)$$

Figure 2 illustrates the model for different choices of Γ^* , where the effect of Γ^* on the shape of the stress-strain curve in the strain softening regime is shown. In this regime, the response depends on the relative dimensions of the specimen to the size of internal defects ([21]). Although the problem of strain localization is not treated in this paper, the present approach has the potential advantage of indirectly capturing the size effect in brittle material response through variation in Γ^* . The curves, in Figure 2, are obtained for the same material parameters used previously for Figure 1 and for η of 0.1. Note that no snap back in the solution is predicted by this method.

Conclusion

The general formulation of elastic-damaging processes for a brittle solid is presented within the framework of the internal theory of thermodynamics. It is shown that the solution to a constraint

damage models. The use of bilinear damage function would lead to an internal contradiction, regardless of the slope used in the strain softening regime.

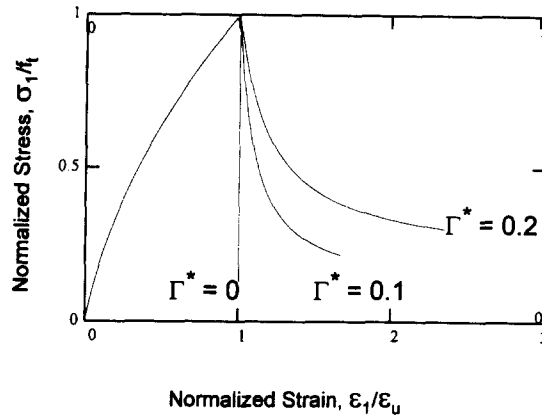


Figure 2. Theoretical stress-strain curves using Equation (18).

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